

CSE321 Quiz 03 Set B

Time 20 minutes

[Q1 5 +Q2 3 +Q3 2]

Q1. You are given a storage system with a total of 1300 blocks. Block size is 128 byte (2^7). The first block is Superblock. The next blocks are : 1 data bitmap block, 1 Inode bitmap block. 50 INODE blocks, and the rest of them are data blocks. The size of Each INODE is 32 byte (2^5). The memory address space to represent one block is 4 bytes(2^2). The INODE structure is: 4 direct blocks, 1 single indirect, 2 double indirect blocks. Given this information,

- What's the maximum size of a file in this inode structure?
- What's the total number of unique INODES that can be stored in this system?
- You are trying to save a folder which has 213 unique files inside [each taking one block]. Will there be any issues? Mathematically prove if yes / no.

Q2. you have opened a directory “/cse321/quiz3”. The file quiz3 has a total of 3 data blocks written to it. You want to append one more data block to it. You have to perform read() on some point of the file system and write() on some to achieve this. [The file is already opened for you]. Write only the correct options among them only.

- read()**: Root INODE, Root Data, cse321 INODE, cse321 Data, quiz3 INODE, quiz3 data[0].quiz3 data[1].quiz3 data[2].quiz3 data[3], INODE bitmap, DataBlock Bitmap.
- write()**: Root INODE, Root Data, cse321 INODE, cse321 Data, quiz3 INODE, quiz3 data[0].quiz3 data[1].quiz3 data[2].quiz3 data[3], INODE bitmap, DataBlock Bitmap.

Q3.

- In the three criteria while writing a data: data bitmap update, inode update, data block write. In which point a crash has to appear for the situation to be labelled as “space leak”?
- In metadata journaling, the actual data is not written in the journal, So data inconsistencies may happen with this form of journaling. What protocol does the OS follow to minimize this? [ONE LINE MAX]